

Temperature_Analysis

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```
rm(list = ls())
```

```
knitr::opts_chunk$set(cache = TRUE, cache.lazy = FALSE, warning = FALSE,  
                        message = FALSE, echo = TRUE, dpi = 300,  
                        fig.width = 15, fig.height = 10)
```

```
library(DT)  
library(tidyverse)  
library(SPEI)  
library(drought)  
library(extRemes)  
library(heatwaveR)
```

Reading the data from the directory

```
## Daily data  
data <- readRDS(file = "/Users/mandarphatak/Documents/GitHub/Climate-Analysis/DWD_Folder/Data_DWD/Hambur
```

data tranformation

```
# conversion to monthly data ( makes it easier for analysis)  
monthly_tbl <- data |>  
  mutate(year = year(date),  
         month = month(date) ) |>  
  group_by(year, month) |>  
  summarize(  
    Tavg = mean(Tavg, na.rm = TRUE),  
    Tmax = mean(Tmax, na.rm = TRUE),  
    Tmin = mean(Tmin, na.rm = TRUE),  
    prec = sum(prec, na.rm = TRUE),  
    rel_hum = mean(rel_hum, na.rm = TRUE),  
  
    .groups = "drop"  
  ) |> drop_na()
```

```
## conversion to annual data (for over all time-period)
annual_tbl <- monthly_tbl |>
  group_by(year) |>
  summarize(
    Tavg = mean(Tavg, na.rm = TRUE),
    Tmax = mean(Tmax, na.rm = TRUE),
    Tmin = mean(Tmin, na.rm = TRUE),
    prec = sum(prec, na.rm = TRUE),
    rel_hum = mean(rel_hum, na.rm = TRUE),
    .groups = "drop"
  )

annual_tbl |> dplyr::select(-year) |> summary()
```

```
##      Tavg      Tmax      Tmin      prec
## Min.   : 6.642   Min.   :10.88   Min.   :1.664   Min.   : 143.0
## 1st Qu.: 8.328   1st Qu.:12.06   1st Qu.:4.593   1st Qu.: 679.2
## Median : 9.099   Median :12.91   Median :5.097   Median : 742.5
## Mean   : 9.033   Mean   :12.89   Mean   :5.098   Mean   : 752.5
## 3rd Qu.: 9.716   3rd Qu.:13.58   3rd Qu.:5.848   3rd Qu.: 829.7
## Max.   :11.180   Max.   :15.07   Max.   :7.161   Max.   :1071.8
##      rel_hum
## Min.   :18.73
## 1st Qu.:78.46
## Median :80.00
## Mean   :79.09
## 3rd Qu.:81.22
## Max.   :84.81
```

```
plot_dir <- "/Users/mandarphatak/Documents/GitHub/Climate-Analysis/DWD_Folder/Plots"
```

Temperature Variation (1936-2025)

```
## Only Temperature
annual_temperature_tbl_long <- annual_tbl |>
  pivot_longer(
    cols = c(Tavg, Tmax, Tmin),
    names_to = "Temperature_Type",
    values_to = "Temperature"
  )
annual_temperature_tbl_long$Temperature_Type <- factor(
  annual_temperature_tbl_long$Temperature_Type,
  levels = c("Tmax", "Tavg", "Tmin"))

trends_tbl <- annual_temperature_tbl_long %>%
  group_by(Temperature_Type) %>%
  summarize(trend = coef(lm(Temperature ~ year))[2]) %>%
  ungroup()
```

```

trends_tbl <- trends_tbl %>%
  mutate(
    label = case_when(
      Temperature_Type == "Tmax" ~ paste0("Maximum Temperature (Trend: +", round(trend, 4), "°C/year)"),
      Temperature_Type == "Tavg" ~ paste0("Average Temperature (Trend: +", round(trend, 4), "°C/year)"),
      Temperature_Type == "Tmin" ~ paste0("Minimum Temperature (Trend: +", round(trend, 4), "°C/year)"),
    ),
    color = case_when(
      Temperature_Type == "Tmax" ~ "#ff1744",
      Temperature_Type == "Tavg" ~ "#fc8d59",
      Temperature_Type == "Tmin" ~ "#4575b4"
    ),
    y = case_when( # adjust based on your plot's Y-axis ranges
      Temperature_Type == "Tmax" ~ 15,
      Temperature_Type == "Tavg" ~ 10.55,
      Temperature_Type == "Tmin" ~ 6.5
    ),
    x = 1938 # horizontal position of the labels
  )

ggplot(annual_temperature_tbl_long, aes(x = year, y = Temperature, colour = Temperature_Type)) +
  geom_line(linewidth = 1) +
  geom_point(aes(fill = Temperature_Type), shape = 21, size = 4, stroke = 1, colour = "white") +
  # Add separate trend lines for each temperature type
  geom_smooth(aes(group = Temperature_Type), method = "lm", se = FALSE, linewidth = 1,
    show.legend = FALSE, color = "black") +
  geom_text(
    data = trends_tbl,
    aes(x = x, y = y, label = label, color = Temperature_Type),
    fontface = "bold", hjust = 0, size = 7, show.legend = FALSE
  ) +

  # Fixed color scales with intuitive temperature colors
  scale_color_manual(
    values = c("Tmax" = "#ff1744", # Hot red for maximum
              "Tavg" = "#fc8d59", # Orange for average
              "Tmin" = "#4575b4"), # Cool blue for minimum
    labels = c("Tmax" = "Maximum Temperature",
              "Tavg" = "Average Temperature",
              "Tmin" = "Minimum Temperature"),
    name = "Temperature Type"
  ) +
  scale_fill_manual(
    values = c("Tmax" = "#ff1744", # Same colors for consistency
              "Tavg" = "#fc8d59",
              "Tmin" = "#4575b4"),
    guide = "none" # Hide fill legend since color legend shows the same info
  ) +

  scale_x_continuous(breaks = c(seq(1936, 2025, 10), 2025),
    expand = c(0.02, 0.02),

```

```

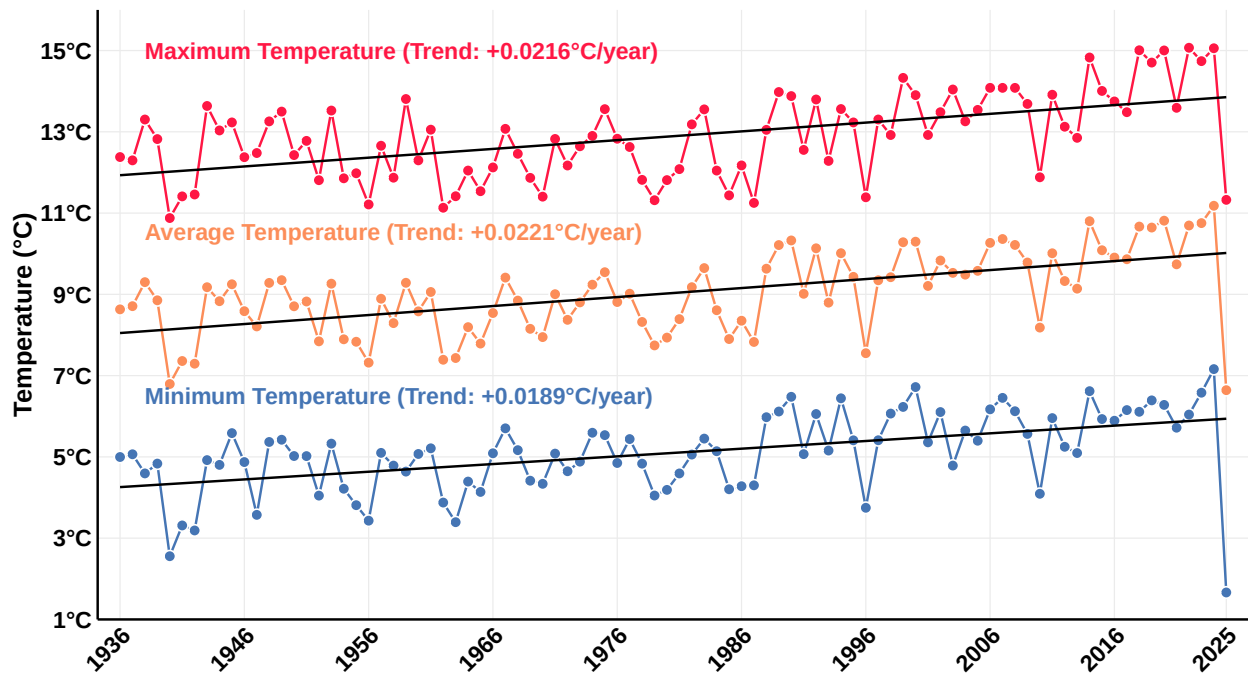
limits = c(1936, 2025)) +
scale_y_continuous(
  limits = c(1, 16),
  breaks = seq(1, 16, by = 2),
  labels = function(x) paste0(format(x, nsmall = 0), "°C"),
  expand = c(0, 0)
) +

labs(
  title = "Temperature Variation (1936-2025)",
  subtitle = "Hamburg Fuhlsbüttel",
  caption = "Source:- Deutscher Wetterdienst (DWD)",
  x = "",
  y = "Temperature (°C)"
) +
theme_minimal() +
theme(
  plot.title.position = "plot",
  plot.title = element_text(hjust = 0, face = "bold", colour = "black", size = 30, margin = margin(b = 10, t = 10, l = 10, r = 10)),
  plot.subtitle = element_text(hjust = 0, face = "bold", colour = "black", size = 30, margin = margin(b = 10, t = 10, l = 10, r = 10)),
  plot.caption = element_text(size = 18, face = "bold", color = "gray50", hjust = 1), #
  axis.line = element_line(color = "black", linewidth = 1),
  axis.title = element_text(face = "bold", color = "black", size = 22),
  axis.ticks = element_blank(),
  panel.grid.minor = element_blank(),
  panel.grid.major = element_line(linewidth = 0.5),
  axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", color = "black", size = 20),
  axis.text.y = element_text(face = "bold", color = "black", size = 20),
  legend.text = element_text(size = 18, face = "bold"),
  legend.title = element_text(size = 18, face = "bold"),
  legend.position = "bottom", # Better positioning for temperature legend
  panel.background = element_blank(),
  plot.background = element_blank()
) +
guides(color = "none", fill = "none")

```

Temperature Variation (1936–2025)

Hamburg Fuhlsbüttel



Source: – Deutscher Wetterdienst (DWD)

```
ggsave(
  filename = file.path(plot_dir, "Annual_Temperature_Trend.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

Decadal Changes in Temperature

```
annual_temperature_tbl_long <- annual_temperature_tbl_long |>
  mutate(decade_group = case_when(
    year >= 1936 & year <= 1939 ~ "1936-1939",
    year >= 1940 & year <= 1949 ~ "1940-1949",
    year >= 1950 & year <= 1959 ~ "1950-1959",
    year >= 1960 & year <= 1969 ~ "1960-1969",
    year >= 1970 & year <= 1979 ~ "1970-1979",
    year >= 1980 & year <= 1989 ~ "1980-1989",
    year >= 1990 & year <= 1999 ~ "1990-1999",
    year >= 2000 & year <= 2009 ~ "2000-2009",
    year >= 2010 & year <= 2019 ~ "2010-2019",
    year >= 2020 & year <= 2025 ~ "2020-2025",
    TRUE ~ NA_character_
  ))
```

```

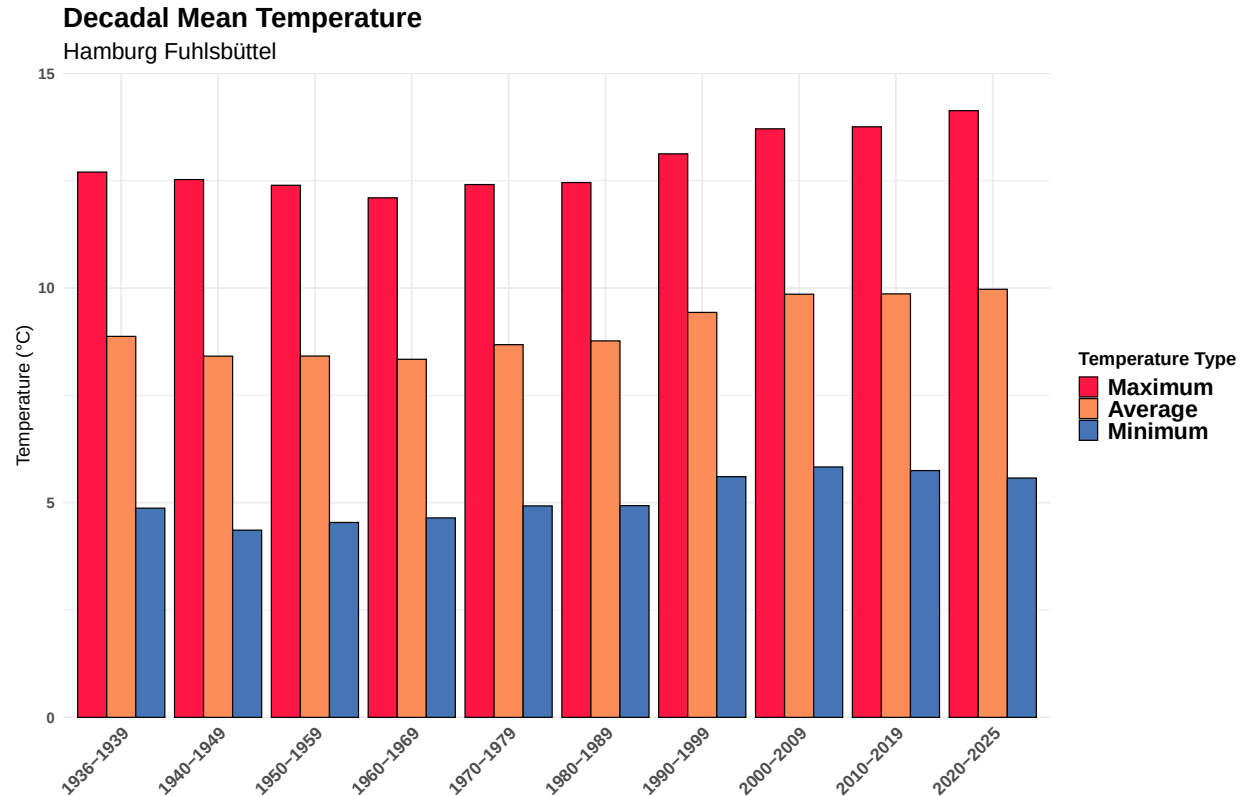
decadal_summary_tbl <- annual_temperature_tbl_long %>%
group_by(decade_group, Temperature_Type) %>%
summarise(decadal_mean = mean(Temperature, na.rm = TRUE), .groups = "drop")

library(ggplot2)

ggplot(decadal_summary_tbl, aes(x = decade_group, y = decadal_mean, fill = Temperature_Type)) +
geom_bar(stat = "identity", position = "dodge", color = "black") +
scale_fill_manual(
  values = c("Tmax" = "#ff1744", "Tavg" = "#fc8d59", "Tmin" = "#4575b4"),
  name = "Temperature Type",
  labels = c("Tmax" = "Maximum", "Tavg" = "Average", "Tmin" = "Minimum")
) +
scale_y_continuous(
  breaks = seq(0, 15, 5), # Customize tick marks
  limits = c(0, 15), # Set axis limits if needed
  expand = c(0, 0) # Remove padding at axis origin
) +
labs(
  title = "Decadal Mean Temperature",
  subtitle = "Hamburg Fuhlsbüttel",

  x = " ",
  y = "Temperature (°C)"
) +
theme_minimal(base_size = 16) +
theme(
  plot.title = element_text(face = "bold", size = 24),
  plot.subtitle = element_text(size = 20),
  axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", size = 15),
  axis.text.y = element_text(face = "bold"),
  legend.title = element_text(face = "bold"),
  legend.text = element_text(face = "bold", size = 20)
)

```



```
ggsave(
  filename = file.path(plot_dir, "Decadal_Temperature_Bar.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

Variation in Temperature across decade

```
ggplot(decadal_summary_tbl, aes(x = decade_group, y = decadal_mean, group = Temperature_Type, color = T
  geom_line(linewidth = 1.5) +
  geom_point(aes(fill = Temperature_Type), size = 8, shape = 21, color = "black", stroke = 1.2) +
scale_color_manual(
  values = c("Tmax" = "#ff1744", "Tavg" = "#fc8d59", "Tmin" = "#4575b4"),
  name = "Temperature Type",
  labels = c("Tmax" = "Maximum", "Tavg" = "Average", "Tmin" = "Minimum")
) +
scale_fill_manual(
  values = c("Tmax" = "#ff1744", "Tavg" = "#fc8d59", "Tmin" = "#4575b4")
) +
scale_y_continuous(
  breaks = c(seq(4, 15, 3), 15), # Customize tick marks
  limits = c(4, 15),           # Set axis limits if needed
  #expand = c(0, 0)             # Remove padding at axis origin
```

```

) +
labs(
  title = "Decadal Temperature Trends (1936-2024)",
  subtitle = "Hamburg Fuhlsbüttel",
  caption = "Source:- Deutscher Wetterdienst (DWD)",
  x = "",
  y = "Temperature (°C)"
) +
theme_minimal(base_size = 16) +
theme(
  plot.title.position = "plot",
  plot.title = element_text(face = "bold", size = 24, margin = margin(b=14)),
  plot.subtitle = element_text(size = 20, margin = margin(b=10)),
  axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", color = "black", size = 20),
  axis.text.y = element_text(face = "bold", size = 20, color = "gray"),
  axis.title.y = element_text(face = "bold", size = 22, color = "gray50"),
  plot.caption = element_text(face = "bold", size = 18, color = "grey"),
  legend.title = element_text(face = "bold"),
  legend.text = element_text(face = "bold"),
  legend.position = "none"

) +

# Add custom legend inside the plot
annotate("point", x = 2.5, y = 13.5, size = 5, shape = 21, fill = "#ff1744") +
annotate("text", x = 2.6, y = 13.5, label = "Maximum Temperature", hjust = 0, size = 6, fontface = "bold")

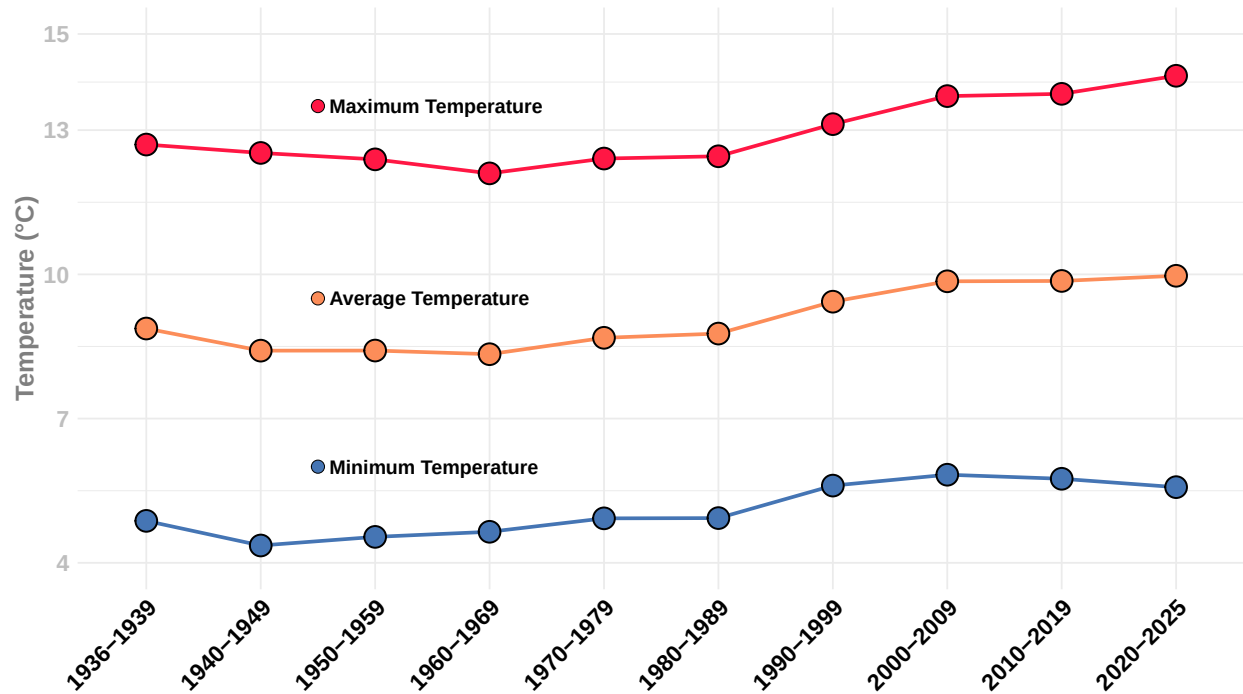
annotate("point", x = 2.5, y = 9.5, size = 5, shape = 21, fill = "#fc8d59") +
annotate("text", x = 2.6, y = 9.5, label = "Average Temperature", hjust = 0, size = 6, fontface = "bold")

annotate("point", x = 2.5, y = 6, size = 5, shape = 21, fill = "#4575b4") +
annotate("text", x = 2.6, y = 6, label = "Minimum Temperature", hjust = 0, size = 6, fontface = "bold")

```

Decadal Temperature Trends (1936–2024)

Hamburg Fuhlsbüttel



Source:– Deutscher Wetterdienst (DWD)

```
ggsave(
  filename = file.path(plot_dir, "Decadal_Temperature_Line.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

Heatwave Events :- Identification of the heatwave events from 1936-2024

```
library(ggrepel) # For repelling labels

identify_heatwaves <- function(date, tmax, threshold, min_duration = 3) {
  # Ensure inputs are valid
  stopifnot(length(date) == length(tmax))
  stopifnot(is.numeric(tmax))
  stopifnot(inherits(date, "Date"))

  # Sort and prepare
```

```

data <- data.frame(date = date, tmax = tmax)
data <- data[order(data$date), ]
data$above_threshold <- data$tmax > threshold

# Initialize output
heatwaves <- data.frame(
  start_date = as.Date(character()),
  end_date = as.Date(character()),
  duration = integer(),
  avg_intensity = numeric(),
  total_intensity = numeric(),
  max_intensity = numeric(),
  mean_tmax = numeric(),
  peak_tmax = numeric(),
  stringsAsFactors = FALSE
)

# Tracking flags
in_heatwave <- FALSE
heatwave_start <- NULL
heatwave_length <- 0

# Loop through
for (i in seq_len(nrow(data))) {
  if (data$above_threshold[i]) {
    if (!in_heatwave) {
      in_heatwave <- TRUE
      heatwave_start <- data$date[i]
      heatwave_length <- 1
    } else {
      heatwave_length <- heatwave_length + 1
    }
  } else {
    if (in_heatwave && heatwave_length >= min_duration) {
      end_date <- data$date[i - 1]
      heatwave_days <- which(data$date >= heatwave_start & data$date <= end_date)
      exceedance <- data$tmax[heatwave_days] - threshold

      heatwaves <- rbind(heatwaves, data.frame(
        start_date = heatwave_start,
        end_date = end_date,
        duration = heatwave_length,
        avg_intensity = mean(exceedance),
        total_intensity = sum(exceedance),
        max_intensity = max(exceedance),
        mean_tmax = mean(data$tmax[heatwave_days]),
        peak_tmax = max(data$tmax[heatwave_days])
      ))
    }
    in_heatwave <- FALSE
    heatwave_length <- 0
  }
}

```

```

# Final check at end
if (in_heatwave && heatwave_length >= min_duration) {
  end_date <- data$date[nrow(data)]
  heatwave_days <- which(data$date >= heatwave_start & data$date <= end_date)
  exceedance <- data$tmax[heatwave_days] - threshold

  heatwaves <- rbind(heatwaves, data.frame(
    start_date = heatwave_start,
    end_date = end_date,
    duration = heatwave_length,
    avg_intensity = mean(exceedance),
    total_intensity = sum(exceedance),
    max_intensity = max(exceedance),
    mean_tmax = mean(data$tmax[heatwave_days]),
    peak_tmax = max(data$tmax[heatwave_days])
  ))
}

# Add year column
if (nrow(heatwaves) > 0) {
  heatwaves$year <- as.numeric(format(heatwaves$start_date, "%Y"))
}

return(heatwaves)
}

heatwave_events_tbl <- identify_heatwaves(
  date = data$date,
  tmax = data$tmax,
  threshold = 28,
  min_duration = 3
)

heatwave_count_per_year <- heatwave_events_tbl %>%
  count(year, name = "num_events")

heatwave_summary_tbl <- heatwave_events_tbl %>%
  group_by(year) %>%
  summarise(
    num_events = n(),
    total_duration = sum(duration, na.rm = TRUE),
    avg_duration = mean(duration, na.rm = TRUE),
    mean_avg_intensity = mean(avg_intensity, na.rm = TRUE),
    mean_max_intensity = mean(max_intensity, na.rm = TRUE),
    total_intensity = sum(total_intensity, na.rm = TRUE),
    max_peak_tmax = max(peak_tmax, na.rm = TRUE),
    .groups = "drop"
  )

# Convert date columns
heatwave_events_tbl$start_date <- as.Date(heatwave_events_tbl$start_date)
heatwave_events_tbl$end_date <- as.Date(heatwave_events_tbl$end_date)

```

```
heatwave_events_tbl$year <- as.numeric(heatwave_events_tbl$year)
#
```

Heatwave events visualization from 1990-2024

```
# Custom color palette
custom_colors <- c("#f8f800", "#fdc70c", "#f3903f", "#ff4d00", "#e60000", "#5e0000")

# Filter events after 1990
heatwave_recent_tbl <- heatwave_events_tbl %>% filter(year >= 1990)

# Count and percentage
recent_events <- nrow(heatwave_recent_tbl)
total_events <- nrow(heatwave_events_tbl)
percent_recent <- round((recent_events / total_events) * 100)

# 95th percentile of avg_intensity in recent events
threshold_95 <- quantile(heatwave_recent_tbl$avg_intensity, 0.95, na.rm = TRUE)

# Max intensity among recent events
max_intensity_value <- max(heatwave_recent_tbl$avg_intensity, na.rm = TRUE)
longest_duration_value <- max(heatwave_recent_tbl$duration, na.rm = TRUE)

# Label most intense event(s)
max_intensity_event <- heatwave_recent_tbl %>%
  filter(avg_intensity == max_intensity_value) %>%
  mutate(label = paste0("Max intensity: ", round(avg_intensity, 1), "\n", format(start_date, "%b %d %Y")))

# Label longest duration (among recent)
longest_duration_event <- heatwave_recent_tbl %>%
  filter(duration == longest_duration_value) %>%
  mutate(label = paste0("Longest: ", duration, " days\n", format(start_date, "%b %d %Y")))

# Extreme events = above 95th percentile (excluding the max to avoid duplicate label)
extreme_events <- heatwave_recent_tbl %>%
  filter(avg_intensity > threshold_95, avg_intensity < max_intensity_value) %>%
  mutate(label = paste0("Extreme: ", round(avg_intensity, 1), "\n", format(start_date, "%b,%d,%Y")))

# Plot
ggplot(heatwave_recent_tbl, aes(x = as.numeric(year), y = avg_intensity)) +
  geom_point(aes(size = duration, color = avg_intensity), alpha = 0.85) +

  geom_smooth(method = "loess", formula = y ~ x, se = FALSE,
    color = "darkblue", linetype = "solid", linewidth = 0.8, alpha = 0.6) +

  geom_hline(yintercept = threshold_95, linetype = "dotted", color = "darkred") +
  annotate("text", x = 1991, y = threshold_95 + 0.09,
    label = "95th Percentile",
    vjust = -0.5, color = "darkred", size = 6) +

  geom_text(data = max_intensity_event, aes(label = label),
    hjust = 1, vjust = 1.2, size = 6, fontface = "bold", color = "#5e0000") +
```

```

ggrepel::geom_text_repel(data = longest_duration_event, aes(label = label),
                        size = 6, fontface = "bold", color = "#5e0000",
                        max.overlaps = 6, box.padding = 0.4, point.padding = 0.4,
                        hjust = 1) +

ggrepel::geom_text_repel(data = extreme_events, aes(label = label),
                        size = 6, fontface = "italic", color = "black",
                        max.overlaps = 10, box.padding = 0.4, point.padding = 0.3,
                        nudge_y = 0.3) +

scale_y_continuous(limits = c(0, 6.25), breaks = seq(0, 6.25, 0.5), expand = c(0, 0.05)) +
scale_x_continuous(limits = c(1990, 2024), breaks = c(seq(1990, 2024, 5), 2024) ) +

scale_color_gradientn(colors = custom_colors,
                      name = "Avg. Intensity",
                      guide = guide_colorbar(barwidth = 1, barheight = 10,
                                              title.position = "top", title.hjust = 0.5,
                                              label.theme = element_text(size = 28))) +

scale_size_continuous(
  range = c(6, 24), # or any range that suits your plot size
  limits = c(3, 12), # ensures 3-12 maps to this range exactly
  breaks = c(3, 6, 9, 12),
  name = "Duration (days)"
) +

guides(
  size = guide_legend(
    title.position = "top",
    title.hjust = 0.5,
    override.aes = list(color = "gray40", fill = "gray40"), # Makes legend dots visible
    label.theme = element_text(size = 28)
  )
) +

labs(
  title = "Extreme Heatwave Events in Hamburg Fuhlsbüttel (1990-2024)",
  subtitle = paste0(
    percent_recent, "% of all heatwaves occurred since 1990"
  ),

  x = "",
  y = "Intensity (°C above threshold)",
  caption = "Source: Deutscher Wetterdienst (DWD)"
) +

theme_minimal(base_size = 15) +
theme(
  plot.title.position = "plot",
  plot.title = element_text(face = "bold", size = 20, margin = margin(b = 15)),
  plot.subtitle = element_text(size = 15, color = "gray30", margin = margin(b = 15), face = "bold"),
  plot.caption = element_text(size = 15, color = "gray50", face = "bold", hjust = 1),
  axis.title.y = element_text(margin = margin(r = 10), face = "bold", size = 22),

```

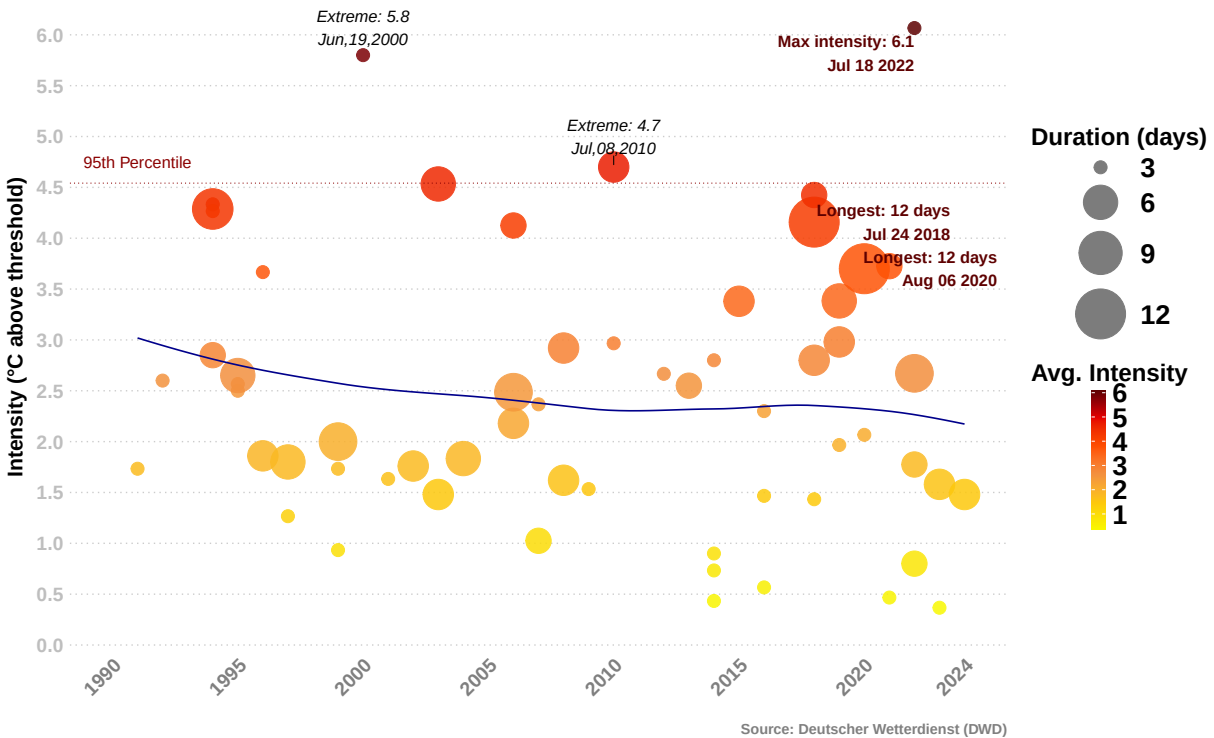
```

axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", color = "gray50", size = 20),
axis.text.y = element_text(face = "bold", color = "gray", size = 20),
legend.position = "right",
legend.box = "vertical",
legend.margin = margin(l = 10, r = 10),
legend.title = element_text(face = "bold", size = 25),
legend.text = element_text(face = "bold", size = 25),
panel.grid.minor = element_blank(),
panel.grid.major.x = element_blank(),
panel.grid.major.y = element_line(color = "gray80", linetype = "dotted"),
plot.margin = margin(t = 20, r = 20, b = 20, l = 20)
)

```

Extreme Heatwave Events in Hamburg Fuhlsbüttel (1990–2024)

57% of all heatwaves occurred since 1990



```

gggsave(
  filename = file.path(plot_dir, "Heatwave_Intensity_Trend_1990_2024.png"),
  plot = last_plot(),
  width = 18,
  height = 12,
  dpi = 450
)

```

Extreme Heatwave events visualization based on max intensity (1990-2024)

```

# Filter post-1990 events
heatwave_recent_tbl <- heatwave_events_tbl %>%
  filter(year >= 1990)

# Total events
recent_events_max <- heatwave_recent_tbl %>%
  filter(max_intensity > 5) %>%
  summarize(count = n()) %>%
  pull(count)

percent_recent_max <- round((recent_events_max / nrow(heatwave_events_tbl)) * 100)

# 95th percentile of max_intensity among recent events
threshold_95_max <- quantile(heatwave_recent_tbl$max_intensity, 0.95, na.rm = TRUE)

# Maximum of max_intensity
max_max_intensity <- max(heatwave_recent_tbl$max_intensity, na.rm = TRUE)

# Max exceedance event
max_max_event <- heatwave_recent_tbl %>%
  filter(max_intensity == max_max_intensity) %>%
  mutate(label = paste0("Max exceedance: ", round(max_intensity, 1), "\n", format(start_date, "%b %d, %Y")))

# Extreme events (excluding absolute max)
extreme_events_max <- heatwave_recent_tbl %>%
  filter(max_intensity > threshold_95_max, max_intensity < max_max_intensity) %>%
  mutate(label = paste0("Extreme: ", round(max_intensity, 1), "\n", format(start_date, "%b %d,%Y")))

# Label longest duration (among recent)
longest_duration_event <- heatwave_recent_tbl %>%
  filter(duration == longest_duration_value) %>%
  mutate(label = paste0("Longest: ", duration, " days\n", format(start_date, "%b %d %Y")))

# Plot for Max Intensity (Post-1990)
ggplot(heatwave_recent_tbl, aes(x = as.numeric(year), y = max_intensity)) +

  geom_point(aes(size = duration, color = max_intensity), alpha = 0.85) +

  geom_smooth(method = "loess", formula = y ~ x, se = FALSE,
    color = "darkblue", linetype = "solid", linewidth = 0.8, alpha = 0.6) +

  geom_hline(yintercept = threshold_95_max, linetype = "dotted", color = "darkred") +
  annotate("text", x = 1995.5, y = threshold_95_max + 0.4, label = "95th Percentile",
    color = "darkred", size = 6) +

  geom_text(data = max_max_event, aes(label = label),
    hjust = 1, vjust = 1.2, size = 6, fontface = "bold", color = "#5e0000") +

  ggrepel::geom_text_repel(data = extreme_events_max, aes(label = label),
    size = 6, fontface = "italic", color = "black",
    max.overlaps = 10, box.padding = 0.4, point.padding = 0.3) +

```

```

scale_y_continuous(limits = c(0, 12), breaks = seq(0, 12, 1), expand = c(0, 0.05)) +
scale_x_continuous(limits = c(1990, 2024), breaks = c(seq(1990, 2024, 5), 2024)) +

scale_color_gradientn(colors = custom_colors,
  name = "Max Intensity",
  guide = guide_colorbar(barwidth = 1.5, barheight = 14,
    title.position = "top", title.hjust = 0.5)) +

  scale_size_continuous(
range = c(6, 24), # or any range that suits your plot size
limits = c(3, 12), # ensures 3-12 maps to this range exactly
breaks = c(3, 6, 9, 12),
name = "Duration (days)"
) +

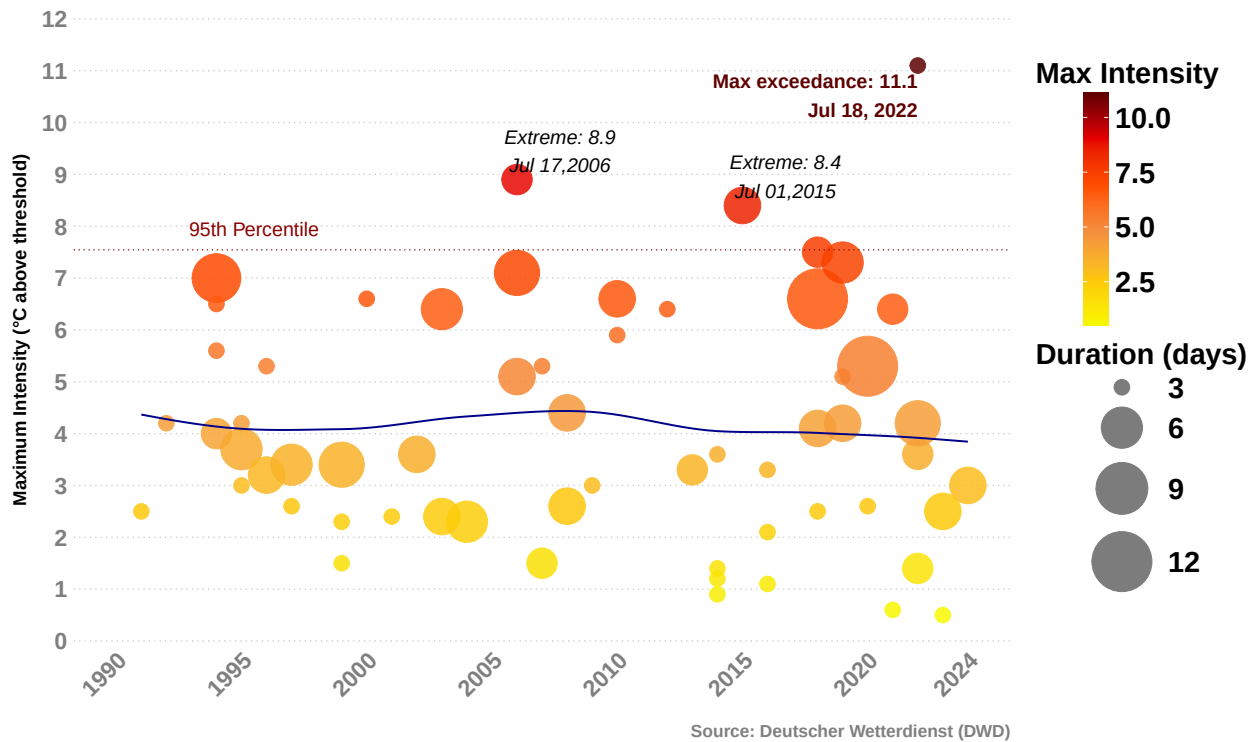
guides(
  size = guide_legend(
    title.position = "top",
    title.hjust = 0.5,
    override.aes = list(color = "gray40", fill = "gray40") # Makes legend dots visible
  )
) + labs(
  title = "Maximum Daily Temperature Exceedance in Hamburg-Fuhlsbüttel Heatwaves (1990-2024)",
  subtitle = paste0("Events exceeding 28°C threshold\n",
    percent_recent_max, "% occurred since 1990"),
  x = "",
  y = "Maximum Intensity (°C above threshold)",
  caption = "Source: Deutscher Wetterdienst (DWD)"
) +
theme_minimal(base_size = 12) +
theme(
  plot.title.position = "plot",
  plot.title = element_text(face = "bold", size = 22, margin = margin(b = 15)),
  plot.subtitle = element_text(size = 15, color = "gray30", margin = margin(b = 15), face = "bold"),
  plot.caption = element_text(size = 15, color = "gray50", face = "bold", hjust = 1),

  axis.title.y = element_text(margin = margin(r = 10), face = "bold", size = 16),
  axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", color = "gray50", size = 20),
  axis.text.y = element_text(face = "bold", color = "gray50", size = 20),
  legend.position = "right",
  legend.box = "vertical",
  legend.margin = margin(l = 10),
  legend.title = element_text(face = "bold", size = 25),
  legend.text = element_text(face = "bold", size = 25),
  panel.grid.minor = element_blank(),
  panel.grid.major.x = element_blank(),
  panel.grid.major.y = element_line(color = "gray80", linetype = "dotted")
)

```

Maximum Daily Temperature Exceedance in Hamburg–Fuhlsbüttel Heatwaves (1990–2024)

Events exceeding 28°C threshold
20% occurred since 1990



```
ggsave(
  filename = file.path(plot_dir, "Heatwave_Max_Exceedance_1990_2024.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

Heat wave magnitude index (hwmid) computation

```
### starting from the data (daily data)
## Daily data
data <- readRDS(file = "/Users/mandarphatak/Documents/GitHub/Climate-Analysis/DWD_Folder/Data_DWD/Hamburg_Heatwaves_1990_2024.rds")
# changing the data variables as in the documentation of of extRemes library, we only focus on Maximum Temperature
data <- data |>
  rename(t = date, Temp = Tmax) |>
  select(t, Temp)
```

Now after we have made the necessary adjustments, so we compute the hwmid as hwmid takes only 32 years for reference which we think will not be justifying the research.

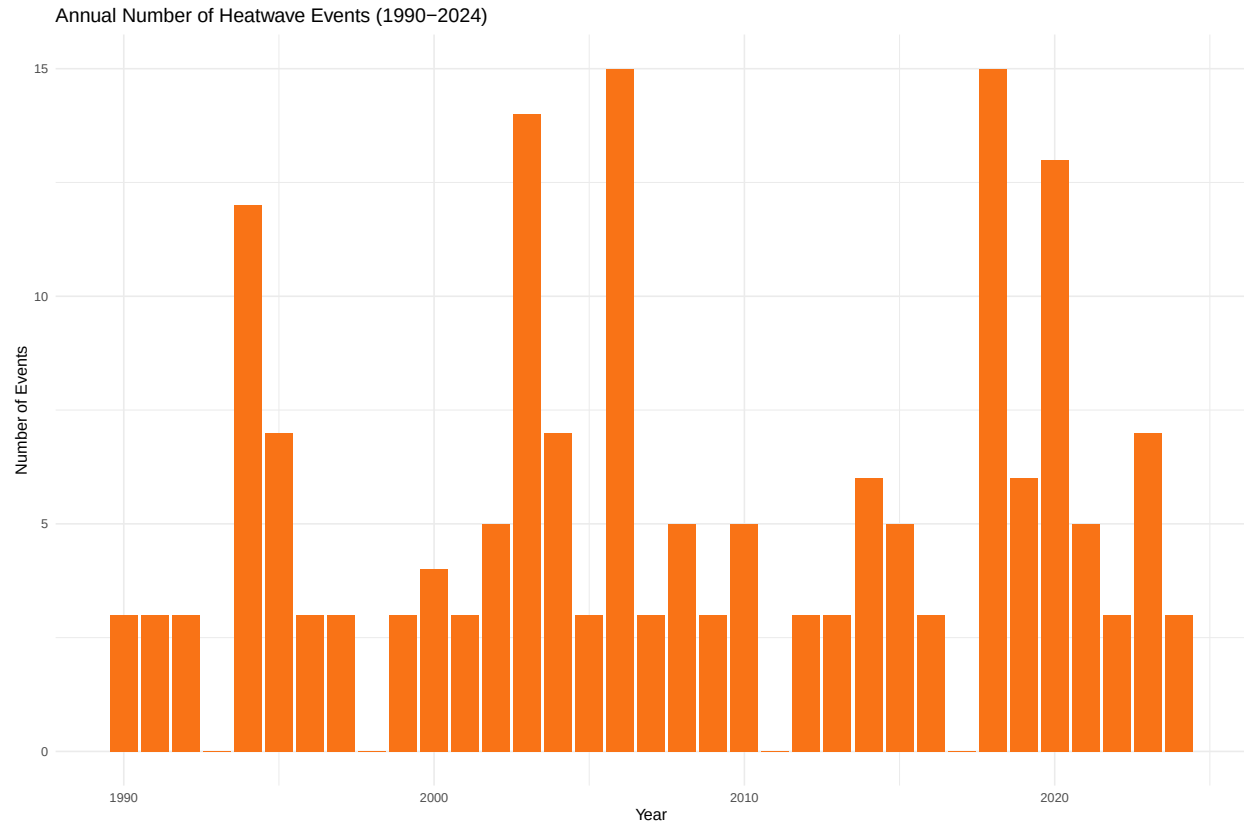
```
# Reference period: 1936-1989 (54 years)
ref_start_year <- 1936
ref_data <- data %>%
  filter(year(t) >= 1936 & year(t) <= 1989)
Tref <- ref_data$Temp

# Analysis period: 1990-2025
analysis_start_year <- 1990
analysis_data <- data %>%
  filter(year(t) >= analysis_start_year)
Temp <- analysis_data$Temp

hwmid_result <- hwmid(yTref = 1936,
  Tref = Tref,
  yTemp = 1990,
  Temp = Temp)

hwmid_tbl <- tibble(
  year = 1990:2024,
  HWMId = hwmid_result$hwmid[,1],
  events = hwmid_result$hwmid[,2],
  peak_day = hwmid_result$hwmid[,3]
)

ggplot(hwmid_tbl, aes(x = year, y = events)) +
  geom_col(fill = "#f97316") +
  #scale_x_continuous(breaks = seq(1990,2024,10), limits = c(1990,2025)) +
  labs(
    title = "Annual Number of Heatwave Events (1990-2024)",
    x = "Year",
    y = "Number of Events"
  ) +
  theme_minimal(base_size = 14)
```



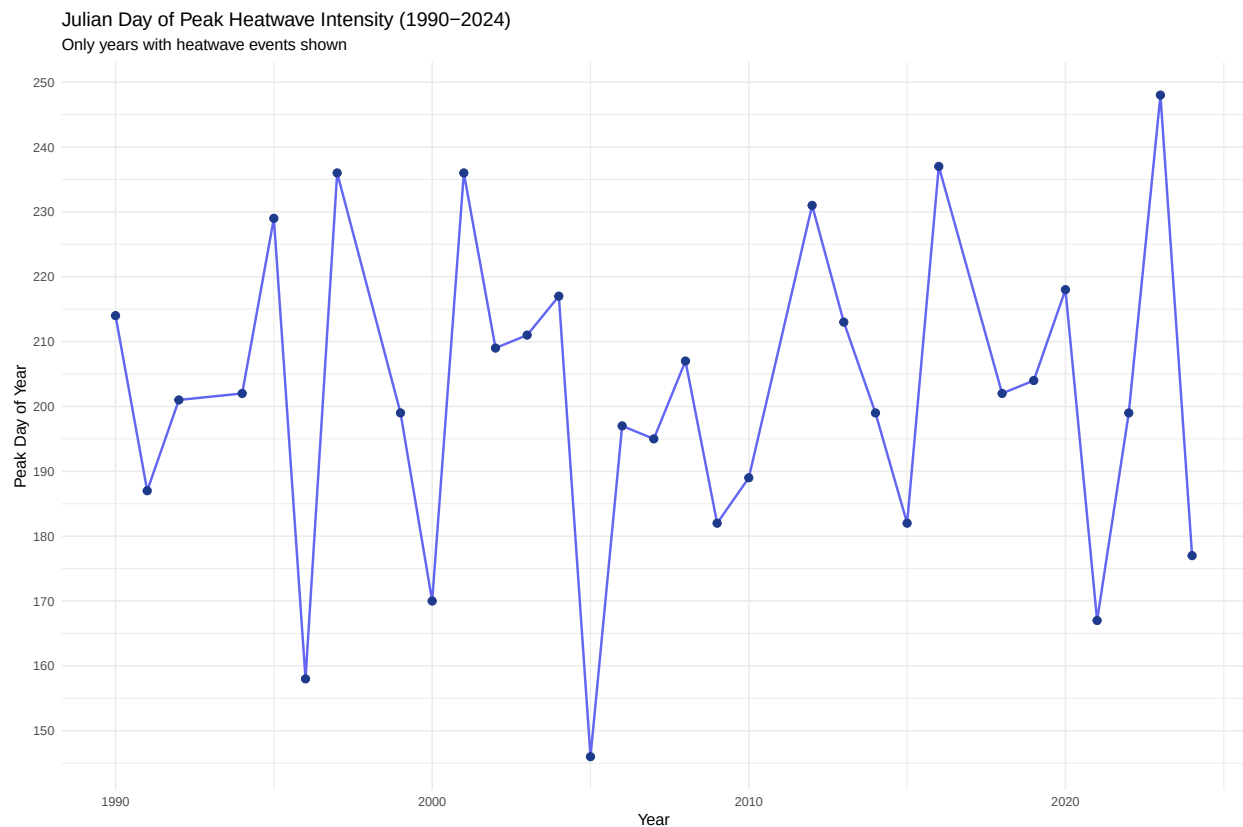
```
ggsave(
  filename = file.path(plot_dir, "HWMId_Annual_Events_1990_2024.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

```
hwmid_tbl <- tibble(
  year = 1990:2024,
  HWMId = hwmid_result$hwmid[,1],
  events = hwmid_result$hwmid[,2],
  peak_day = hwmid_result$hwmid[,3]
)

# Filter only years with actual heatwave events
peak_day_tbl <- hwmid_tbl %>% filter(peak_day > 0)

ggplot(peak_day_tbl, aes(x = year, y = peak_day)) +
  geom_line(color = "#6366f1", linewidth = 1) +
  geom_point(color = "#1e3a8a", size = 3) +
  labs(
    title = "Julian Day of Peak Heatwave Intensity (1990-2024)",
    subtitle = "Only years with heatwave events shown",
    x = "Year",
    y = "Peak Day of Year"
  ) +
```

```
scale_y_continuous(breaks = seq(150, 250, 10)) +
theme_minimal(base_size = 14)
```



```
ggsave(
  filename = file.path(plot_dir, "HWMId_Peak_Julian_Day_Line.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

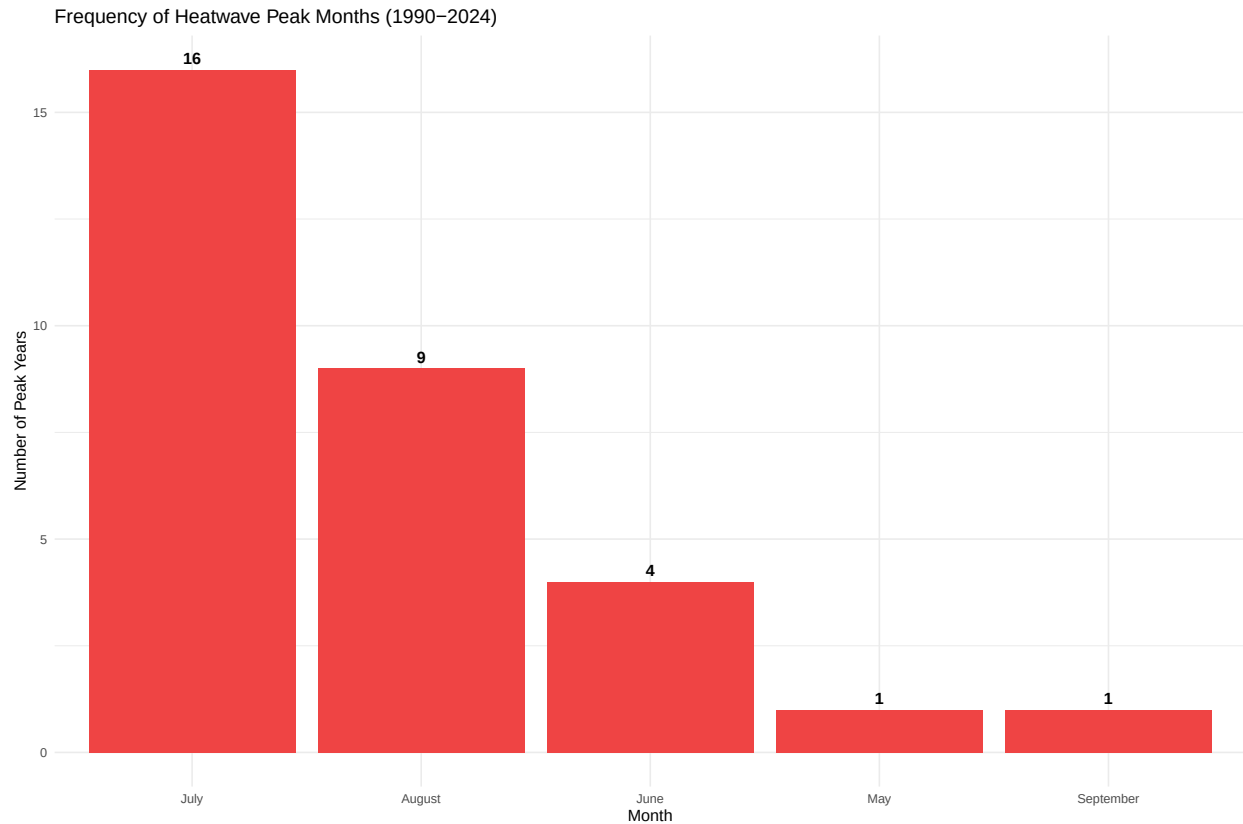
```
# Filter valid peak days (non-zero)
peak_day_tbl <- hwmid_tbl %>%
  filter(peak_day > 0) %>%
  mutate(
    peak_date = as.Date(peak_day - 1, origin = paste0(year, "-01-01")),
    peak_month = format(peak_date, "%B"), # Full month name
    peak_day_of_month = format(peak_date, "%d") # Day of the month (01-31)
  )
```

```
peak_day_tbl %>%
  count(peak_month) %>%
  ggplot(aes(x = reorder(peak_month, -n), y = n)) +
  geom_col(fill = "#ef4444") +
  geom_text(aes(label = n), vjust = -0.5, fontface = "bold", size = 5) +
  labs(
```

```

title = "Frequency of Heatwave Peak Months (1990-2024)",
x = "Month",
y = "Number of Peak Years"
) +
theme_minimal(base_size = 14)

```



```

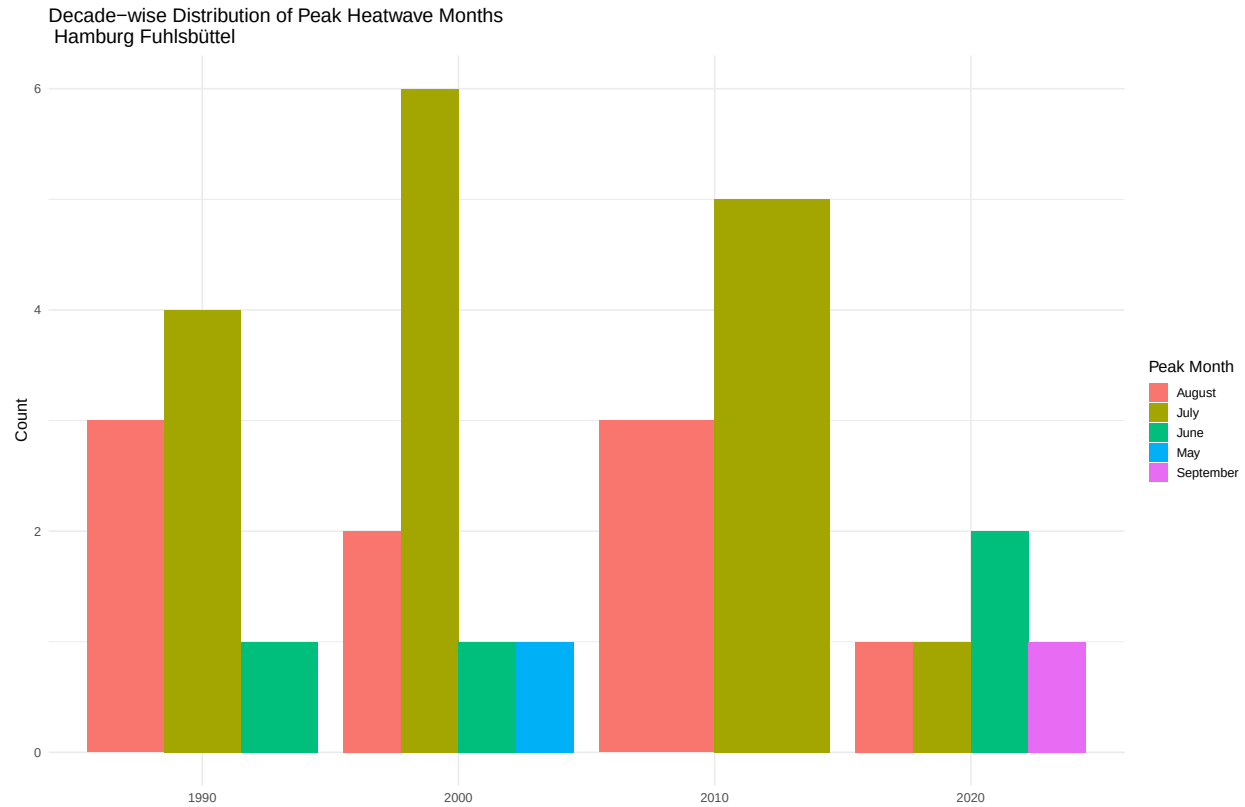
ggsave(
  filename = file.path(plot_dir, "Heatwave_Peak_Month_Count.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)

```

```

peak_day_tbl %>%
  mutate(decade = floor(year / 10) * 10) %>%
  count(decade, peak_month) %>%
  ggplot(aes(x = factor(decade), y = n, fill = peak_month)) +
  geom_col(position = "dodge") +
  labs(
    title = "Decade-wise Distribution of Peak Heatwave Months\n Hamburg Fuhlsbüttel",
    x = "",
    y = "Count",
    fill = "Peak Month"
  ) +
  theme_minimal(base_size = 14)

```



```
ggsave(
  filename = file.path(plot_dir, "Peak_Months_by_Decade.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

Heat Wave Magnitude Index

```
library(heatwaveR)

## Daily data
data <- readRDS(file = "/Users/mandarphatak/Documents/GitHub/Climate-Analysis/DWD_Folder/Data_DWD/Hamburg")

# modifications as in the documentation
data_tbl <- data |>
  rename(t = date) |>
  select(t, temp = Tmax)

# heat Wave Frequency per decade
res_1936_1989 <- detect_event(ts2clm(data = data_tbl,
  climatologyPeriod = c("1936-01-01", "1989-12-31")))
```

```

res_1990_2025 <- detect_event(ts2clm(data = data_tbl,
                                     climatologyPeriod = c("1990-01-01", "2025-05-28")))

res_1936_1989$event <- res_1936_1989$event |>
  select(date_start, date_peak, date_end, duration, intensity_mean, intensity_max) |>
  arrange(desc(intensity_max))

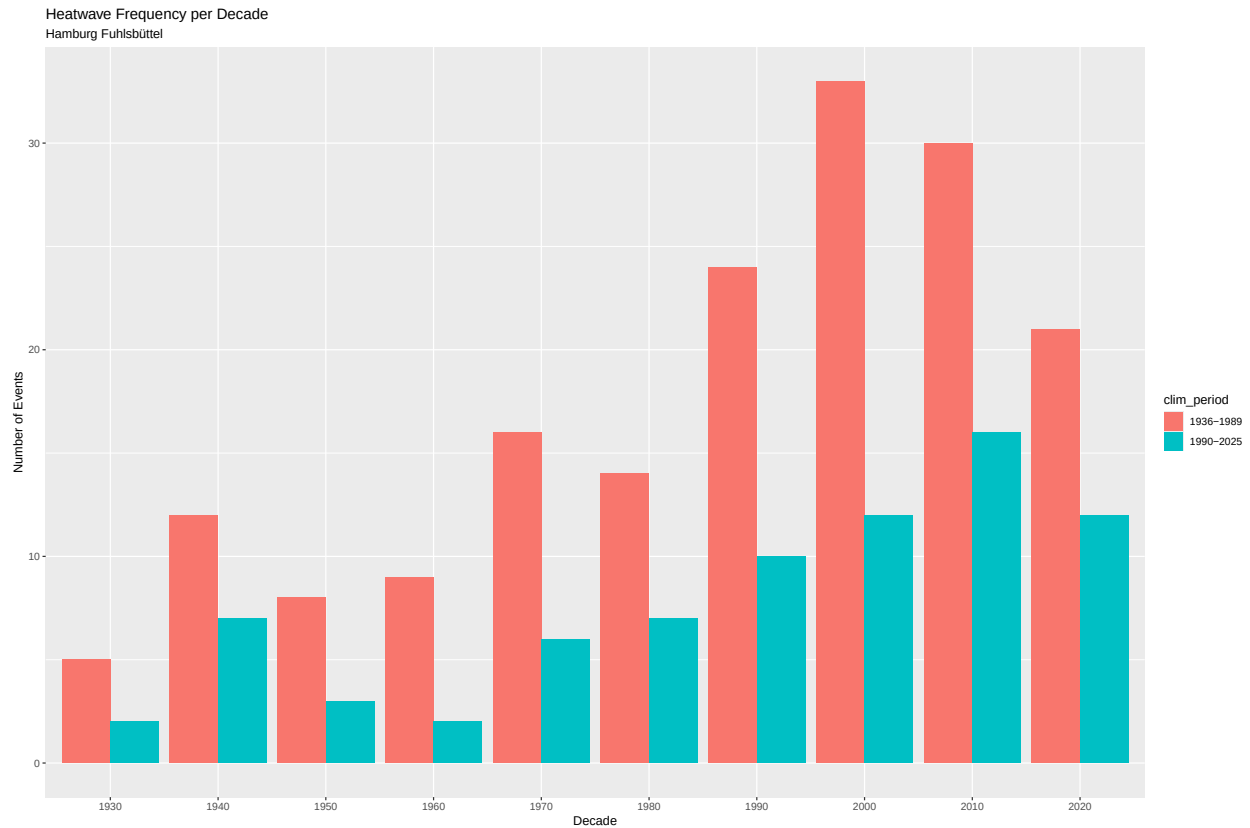
res_1990_2025$event <- res_1990_2025$event |>
  select(date_start, date_peak, date_end, duration, intensity_mean, intensity_max) |>
  arrange(desc(intensity_max))

events_1936_1989 <- res_1936_1989$event |>
  mutate(clim_period = "1936-1989")

events_1990_2025 <- res_1990_2025$event |>
  mutate(clim_period = "1990-2025")
combined_events <- bind_rows(events_1936_1989, events_1990_2025)

## We can clearly see that using the refrence period has an impact on the heatwave events
combined_events %>%
  mutate(year = year(date_start), decade = floor(year / 10) * 10) %>%
  group_by(decade, clim_period) %>%
  summarise(
    frequency = n(),
    avg_intensity = mean(intensity_mean),
    max_intensity = max(intensity_max),
    .groups = "drop"
  ) %>%
  ggplot(aes(x = factor(decade), y = frequency, fill = clim_period)) +
  geom_col(position = "dodge") +
  labs(
    title = "Heatwave Frequency per Decade",
    subtitle = "Hamburg Fuhlsbüttel",
    y = "Number of Events", x = "Decade"
  )

```



```
ggsave(
  filename = file.path(plot_dir, "Heatwave_Frequency_Climatology_Period.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)
```

Exceedance

```
# Use full period if analyzing long-term trends
exceed_28c <- exceedance(
  data = data_tbl,
  x = t,
  y = temp,
  threshold = 28,
  below = FALSE,
  minDuration = 3,
  joinAcrossGaps = TRUE, # or FALSE if you don't want to link across any gaps
  maxGap = 0,
  roundRes = 4,
  returnDF = TRUE
)
```

```

exceed_28c_events <- exceed_28c$exceedance %>%
  mutate(
    year = year(date_start),
    month = month(date_start, label = TRUE, abbr = TRUE)
  )

```

```

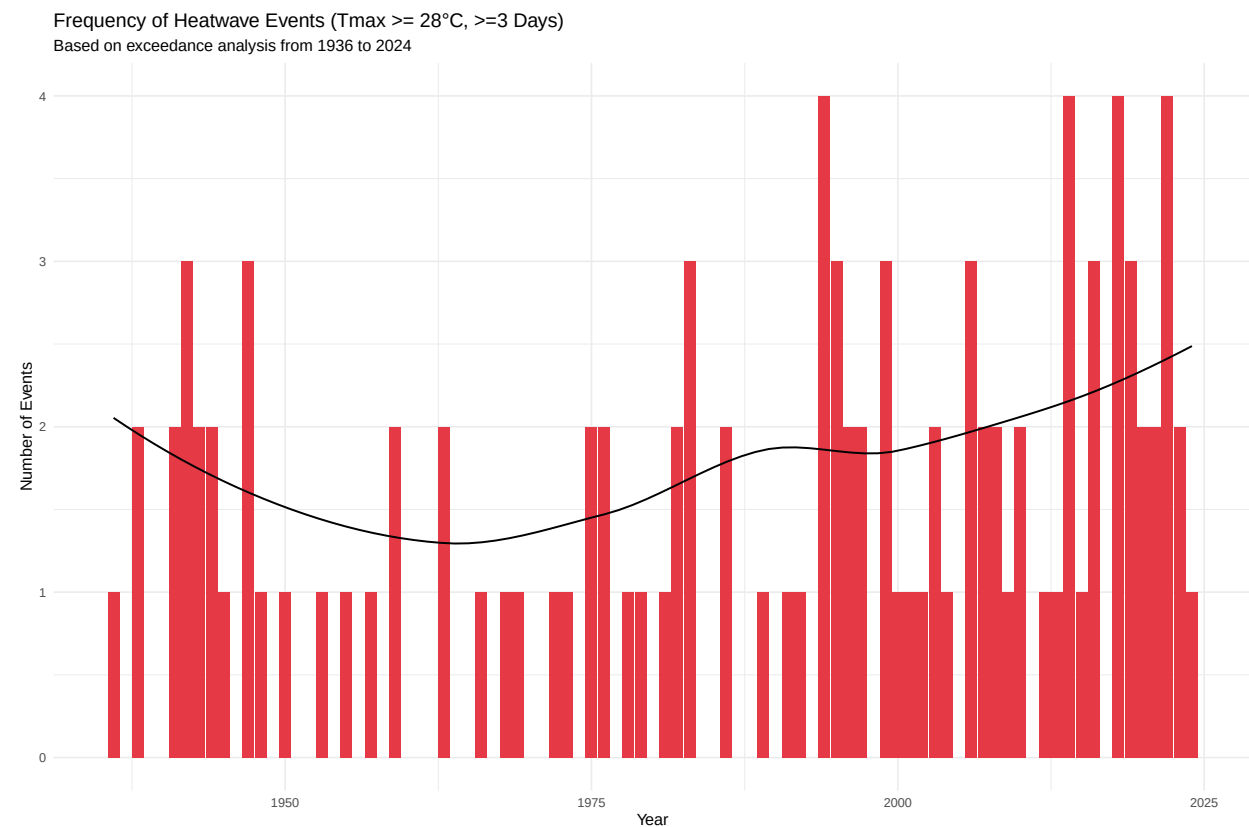
library(ggplot2)

```

```

exceed_28c_events %>%
  count(year) %>%
  ggplot(aes(x = year, y = n)) +
  geom_col(fill = "#e63946") +
  geom_smooth(method = "loess", se = FALSE, color = "black", linewidth = 0.8) +
  labs(
    title = "Frequency of Heatwave Events (Tmax >= 28°C, 3 Days)",
    subtitle = "Based on exceedance analysis from 1936 to 2024",
    x = "Year", y = "Number of Events"
  ) +
  theme_minimal(base_size = 14)

```



```

ggsave(
  filename = file.path(plot_dir, "Exceedance_28C_Frequency_Annual.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
)

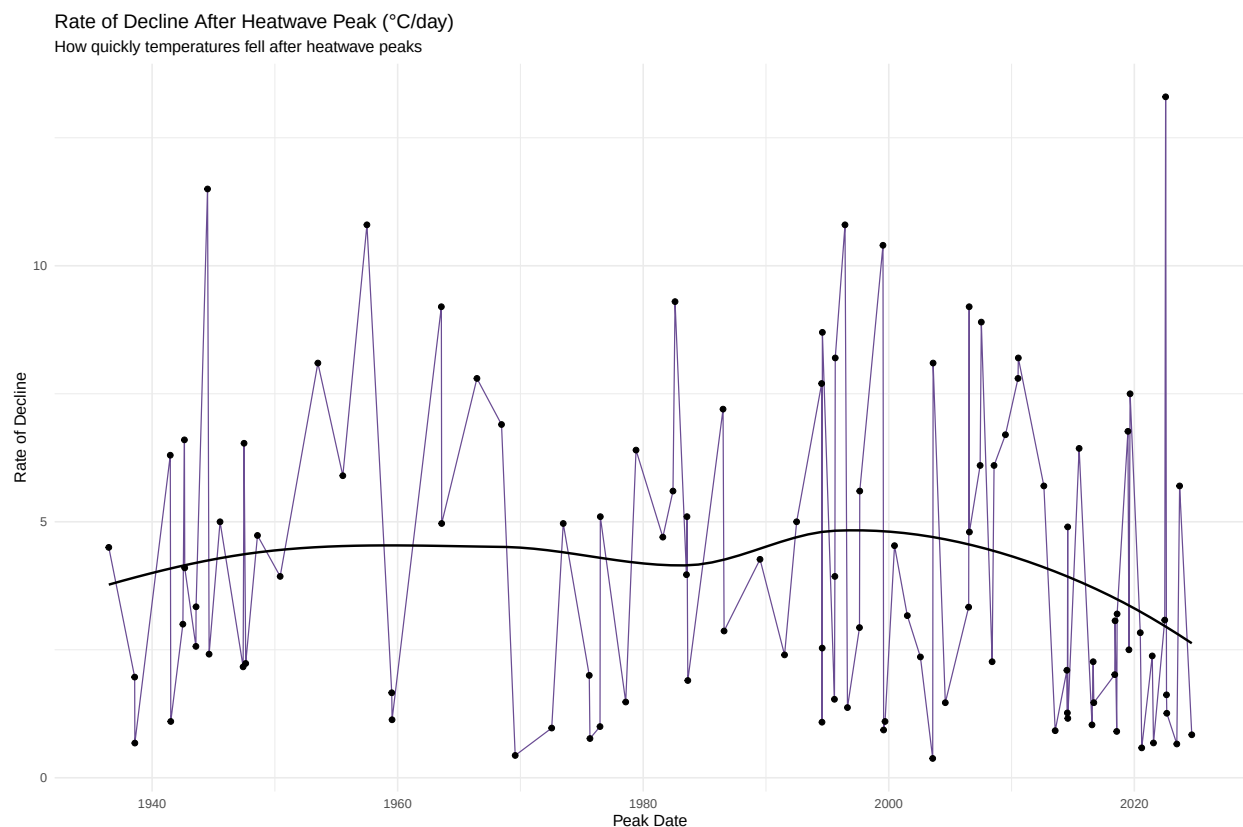
```

```

    dpi = 450
  )

ggplot(exceed_28c_events, aes(x = date_peak, y = rate_decline)) +
  geom_line(color = "#6a4c93") +
  geom_point(size = 2) +
  geom_smooth(method = "loess", se = FALSE, color = "black") +
  labs(
    title = "Rate of Decline After Heatwave Peak (°C/day)",
    x = "Peak Date", y = "Rate of Decline",
    subtitle = "How quickly temperatures fell after heatwave peaks"
  ) +
  theme_minimal(base_size = 14)

```



```

ggsave(
  filename = file.path(plot_dir, "Exceedance_Rate_Decline.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)

```

```

# Find top 5 slowest and fastest declines
slow_decline <- exceed_28c_events %>% arrange(rate_decline)
fast_decline <- exceed_28c_events %>% arrange(desc(rate_decline))

```

```

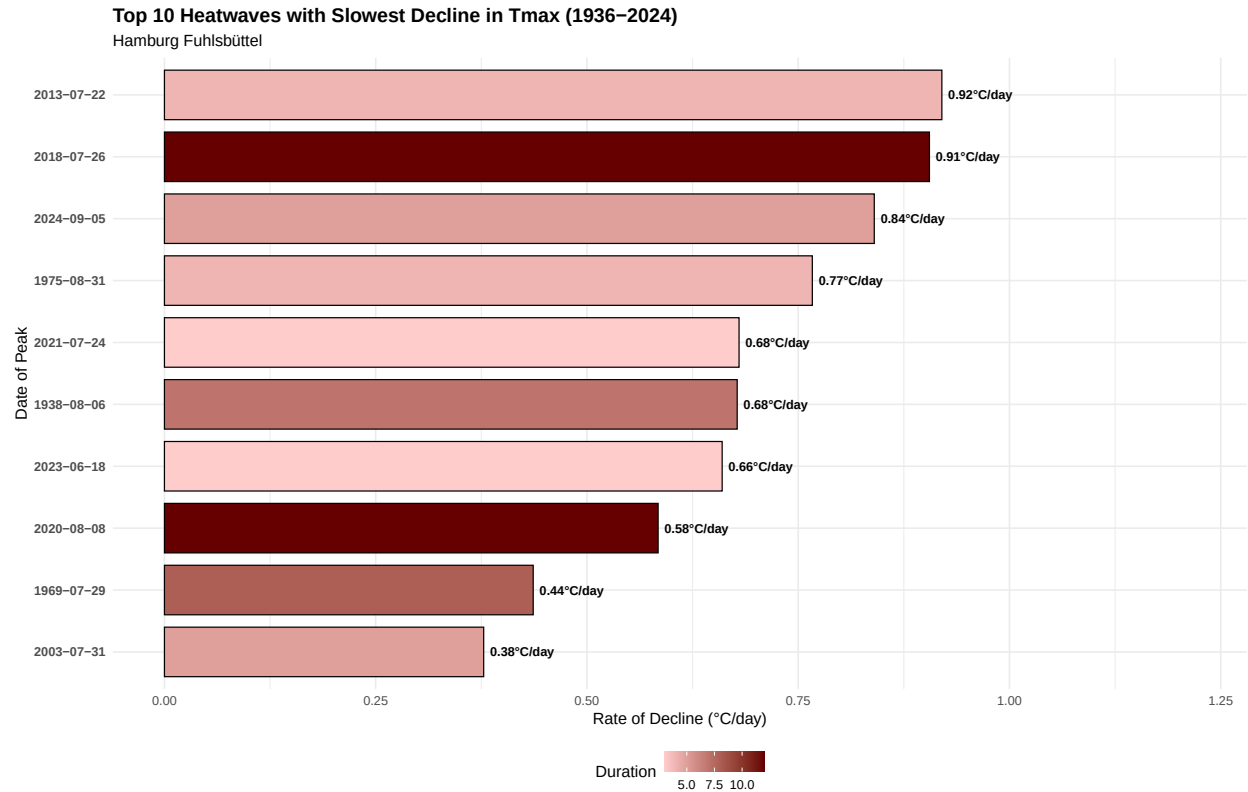
slow_decline_events <- exceed_28c_events %>%
  arrange(rate_decline) %>%
  slice(1:10) %>%
  mutate(label = paste0("Slow Decline: ", round(rate_decline, 2), "°C/day\n", format(date_start, "%b %d

library(ggplot2)
library(scales)

# Filter top 10 slowest rate of decline
top10_decline <- exceed_28c$exceedance %>%
  arrange(rate_decline) %>%
  slice(1:10) %>%
  mutate(label = paste0(date_peak, "\n", sprintf("%.2f°C/day", rate_decline)))

# Plot
ggplot(top10_decline, aes(x = rate_decline, y = reorder(as.character(date_peak), rate_decline))) +
  geom_col(aes(fill = duration), color = "black", width = 0.8) +
  geom_text(aes(label = sprintf("%.2f°C/day", rate_decline)),
            hjust = -0.1, size = 4, fontface = "bold") +
  scale_fill_gradient(low = "#ffcccc", high = "#660000", name = "Duration") +
  labs(
    title = "Top 10 Heatwaves with Slowest Decline in Tmax (1936-2024)",
    subtitle = "Hamburg Fuhlsbüttel",
    caption = "Source:- Deutscher Wetterdienst (DWD)",
    x = "Rate of Decline (°C/day)",
    y = "Date of Peak"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(face = "bold"),
    legend.position = "bottom",
    plot.caption = element_text(size = 15, color = "gray50", face = "bold", hjust = 1)
  ) +
  coord_cartesian(xlim = c(0, max(top10_decline$rate_decline) + 0.3))

```



```
ggsave(
  filename = file.path(plot_dir, "Top10_Slowest_Decline_Rate.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)

# Filter top 10 by duration
top10_duration <- exceed_28c$exceedance %>%
  arrange(desc(duration)) %>%
  slice(1:10) %>%
  mutate(label = paste0(date_peak, "\n", duration, " days"))

# Plot
ggplot(top10_duration, aes(x = duration, y = reorder(as.character(date_peak), duration))) +
  geom_col(aes(fill = rate_decline), color = "black", width = 0.8) +
  geom_text(aes(label = paste0(duration, " days")),
    hjust = -0.1, size = 4, fontface = "bold") +
  scale_fill_gradientn(
    colours = c("#cce5ff", "#66a3cc", "#004c99", "#003366"), # light to dark
    values = scales::rescale(c(0.4, 1.5, 3.5, 9.3)), # breakpoints
    name = "Rate of Decline"
  ) +
  labs(
    title = "Top 10 Longest Duration Heatwave Events (1936–2024)",
    subtitle = "Hamburg Fuhlsbüttel",
  )
```

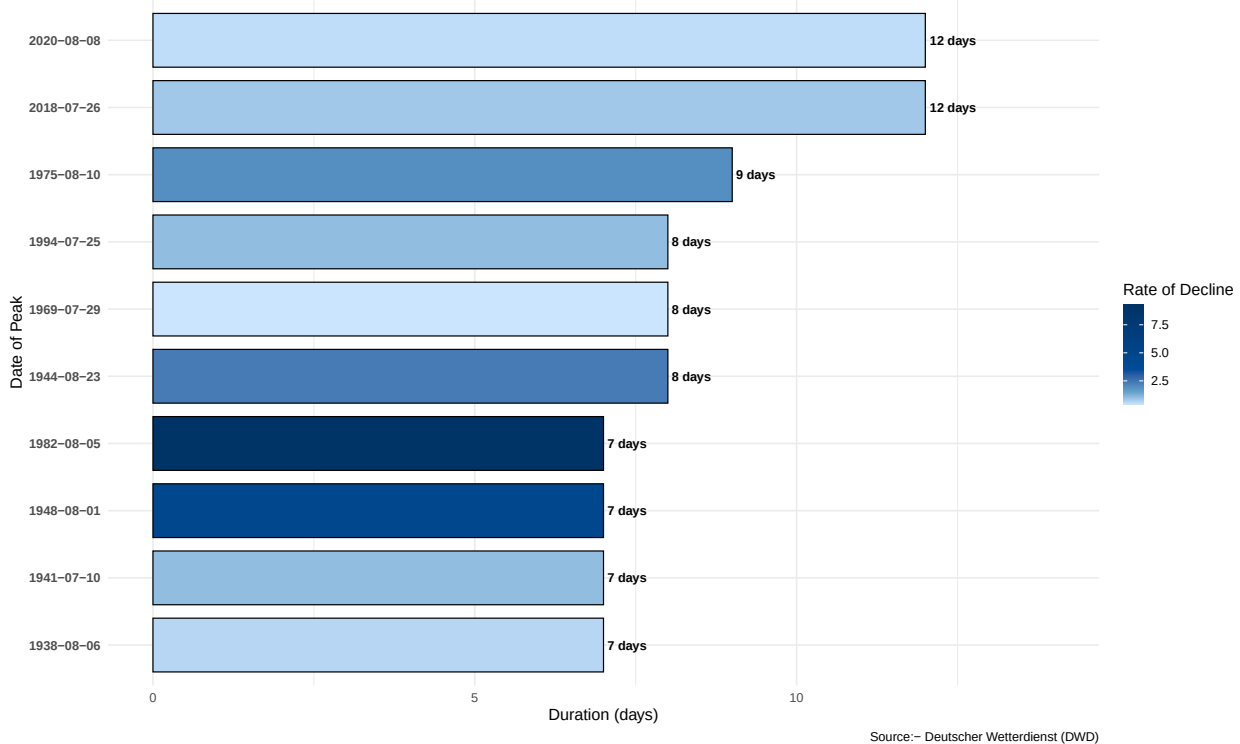
```

caption = "Source:- Deutscher Wetterdienst (DWD)",
x = "Duration (days)",
y = "Date of Peak"
) +
theme_minimal(base_size = 14) +
theme(
  plot.title.position = "plot",
  plot.title = element_text(face = "bold", size = 22, margin = margin(b=15)),
  plot.subtitle = element_text(face = "bold", size = 18, margin = margin(b=10)),
  axis.text.y = element_text(face = "bold"),
  legend.position = "right"
) +
coord_cartesian(xlim = c(0, max(top10_duration$duration) + 2))

```

Top 10 Longest Duration Heatwave Events (1936–2024)

Hamburg Fuhlsbüttel



```

ggsave(
  filename = file.path(plot_dir, "Top10_Longest_Heatwaves.png"),
  plot = last_plot(),
  width = 15,
  height = 10,
  dpi = 450
)

```

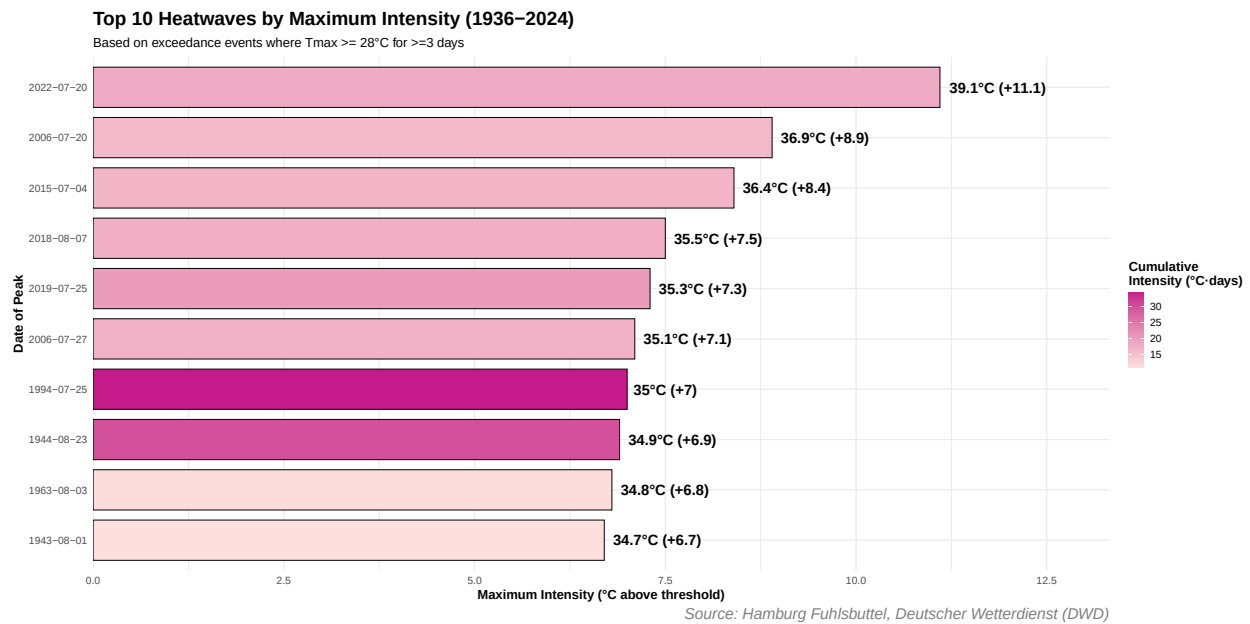
“Among the top 10 longest-duration heatwave events from 1936–2024, those with the longest persistence (e.g., 2020 and 2018) exhibited low rates of decline, suggesting prolonged high-temperature stress with slow recovery. In contrast, shorter events such as in 1982, despite lasting only 7 days, showed much steeper decline rates, indicating a more abrupt end to extreme heat conditions. This distinction has implications for

understanding not only intensity but also post-heatwave cooling dynamics, relevant for public health and infrastructure recovery.”

```
library(ggplot2)
library(dplyr)

# Select top 10 by intensity_max
top_max_intensity <- exceed_28c$exceedance %>%
  arrange(desc(intensity_max)) %>%
  slice_head(n = 10) %>%
  mutate(
    actual_tmax = round(intensity_max + 28, 1),
    label = paste0(actual_tmax, "°C (+", round(intensity_max, 1), "°C)",
    date_peak = as.Date(date_peak)
  )

# Plot
# Plot
ggplot(top_max_intensity, aes(x = reorder(format(date_peak, "%Y-%m-%d"), intensity_max),
                             y = intensity_max,
                             fill = intensity_cumulative)) +
  geom_col(color = "black", width = 0.8) +
  geom_text(aes(label = label),
            hjust = -0.1, color = "black", fontface = "bold", size = 6) + # + increased text size
  scale_fill_gradient(low = "#fde0dd", high = "#c51b8a",
                     name = "Cumulative\nIntensity (°C·days)") +
  scale_y_continuous(expand = expansion(mult = c(0, 0.2))) +
  coord_flip() +
  labs(
    title = "Top 10 Heatwaves by Maximum Intensity (1936-2024)",
    subtitle = "Based on exceedance events where Tmax ≥ 28°C for 3 days",
    x = "Date of Peak",
    y = "Maximum Intensity (°C above threshold)",
    caption = "Source: Hamburg Fuhsbuttel, Deutscher Wetterdienst (DWD)"
  ) +
  theme_minimal(base_size = 15) +
  theme(
    plot.title = element_text(face = "bold", size = 22, color = "black", margin = margin(b=10)),
    plot.subtitle = element_text(size = 15, margin = margin(b=8)),
    axis.text.y = element_text(size = 12),
    axis.title.x = element_text(size = 15, face = "bold", color = "black"),
    axis.title.y = element_text(size = 15, face = "bold", color = "black"),
    legend.title = element_text(face = "bold"),
    legend.text = element_text(size = 12),
    plot.caption = element_text(size = 18, face = "italic", colour = "gray50")
  )
```



```
ggsave(  
  filename = file.path(plot_dir, "Top10_Max_Intensity_Events.png"),  
  plot = last_plot(),  
  width = 20,  
  height = 10,  
  dpi = 450  
)
```